ATLAS proposal fact sheet

Title: **Configuration mixing in 0+ and 2+ states in 32Si**

Proposal type: **New proposal**

Part of thesis: **No**

Principal investigator (PI) and Co-PI contact information

PI: **Matthew Martin - matthew.martin@anl.gov**
Co-PI: **Not specified**

Beam time

Days requested: **3**
Tuning included: **No**
Total days approved: **See letter**

Continuous run: **Yes**
Time splitting: **-**
Days between runs: **-**

Beam specifications

Type: **In-flight (RAISOR)**

Type	Species	Energy	Intensity	Purity (%)
In-flight	**31Si**	**10 MeV/u**	**4.00E+05 pps**	**5**
Primary	**TBD**	**TBD MeV/u**	**TBD pnA**	**N/A**
Add'l stable	**none**	**-**	**-**	**N/A**

Special beam timing devices

PII sweeper: **No**

Rebuncher: **Yes**

RIB sweeper: **Yes**

End station

HELIOS (A5)

Targets and beam stop materials

Targets: **CD2**
Thickness (mg/cm2): **0.1**
General target specs: **is provided by CATS or ATLAS**
Beam stop: **Not applicable**
Production target (for in-flight beam): **TBD**

Radiation safety

Calibration sources (owned by ATLAS/PHY): **Alpha**
Using external radioactive materials (ERM): **No**
Type of ionizing radiation of ERM: **-**
Total activity of ERM (Bq): **-**
ERM description: **-**

Electrical safety

Using external electrical equipment: **No**
Maximum voltage (V): **-**
External electrical equipment description: **-**

General safety

Using flammable gases: **No**
Lifting heavy equipment: **No**
Unusual operations/requirements: **N/A**

Certification of information: The PI has certified that the information presented on this form is correct and that all of the collaborators listed on the attached proposal are fully aware of the proposal and have agreed to participate in the experiment.

Configuration mixing in 0^+ and 2^+ states in ^{32}Si

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The onset of the $N = 20$ island of inversion can provide insight into how the fundamental interactions which govern nuclei manifest at the scale of nuclear observables. The structure of ^{32}Si, lying at the edge of this region, has been of significant interest recently and is predicted to have a pure *sd*-shell ground state, with little-to-no pf-shell configurations at low-energy states. This is in contrast to the neighbouring ^{34}Si, where both the 2_1^+ and 0_2^+ states are shown to have dominating pf-shell contributions. The proposed work is to measure cross-sections of low-energy states in ^{32}Si using the ^{31}Si(d,p) reaction and HELIOS to extract the relative *sd*- and pf-shell contributions to the two lowest 0^+ and 2^+ states. We are requesting three days of ^{31}Si beam to perform this measurement.

I. MOTIVATION

In several neutron rich regions of the nuclear chart, it has been observed that the shell closures predicted by the spherical shell model change (e.g. Refs. [1, 2]). Thus, there exist nuclei in which the ground state contains significant contributions from *inverted* configurations, or those with multi-particle multi-hole excitations compared to the nominal spherical shell structure. The first of these regions to be discovered is centred around ^{32}Mg ($N = 20$) [3], and is referred to as the $N = 20$ island of inversion. Further research has demonstrated that the ground state of ^{32}Mg has significant contributions from pf-shell states [4], and is actually dominated by the 2p2h and 4p4h configurations over the nominal 0p0h [5]. On either side of ^{32}Mg, the ground states of ^{31,33}Mg have also been shown to have significant intruder configurations [6].

Moving away from the centre of the island of inversion at ^{32}Mg, intruder configurations are seen in low-energy states, such as the first excited 0^+ states in ^{30}Mg [7] and ^{34}Si [8], rather than in the ground state. Even further from ^{32}Mg, intruder configurations play an

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important role in the intermediate-energy structure of nuclei, such as the spin-trapped 5^- isomer in ^{32}Si [9].

To fully quantify the extent of the $N = 20$ island of inversion and understand the underlying interactions, measurements on nuclei in the vicinity of ^{32}Mg are essential. In the last few years alone, many studies have been performed on the surrounding nuclei, including measurements of ^{29}Na (e.g. Ref. [10]), ^{28}Mg (e.g. Refs. [11–13]), ^{29}Mg (e.g. Ref. [14]), ^{31}Mg (e.g. Ref. [15]), ^{36}Al (e.g. Ref. [16]), ^{32}Si (e.g. Refs. [9, 17]), and ^{34}Si (e.g. Ref. [18]).

The recent work on ^{32}Si has been of much interest when considering intruder configurations and cross-shell excitations in the intermediate-energy states. Prior to a recent measurement [9], the nature of a 5^- nanosecond isomer was debated: one study suggested that the isomer lay above the 4^+ state and that the small decay energy resulted in the isomerism [19], while another placed the 5^- state lower in energy than the 4^+ and the isomerism was caused by the spin change for the $5^- \rightarrow 2^+$ transition [20]. Williams *et al.* confirmed that the 5^- state lies below the 4^+ state and that it is a "spin-trapped" isomer. Importantly for the $N = 20$ island of inversion, the 5^- state was shown to contain dominating $(\nu 0d_{3/2})^{-1} \otimes (\nu 0f_{7/2})^1$ configurations [17], with $\nu 0f_{7/2}$ occupation numbers of 0.92 and 0.96 from the FSU [21] and SPDF-MU [22] shell model calculations. Furthermore, a recent experiment performed using HELIOS at ATLAS has confirmed substantial occupation of the $\nu 0f_{7/2}$ orbital [23], though analysis of these data is ongoing.

Closer to the centre of the $N = 20$ island of inversion and at lower energy, both the $0^+_{\text{g.s.}}$ and 2^+_1 states in ^{34}Si were found to have significant configuration mixing between *sd*- and *pf*-shell orbitals, with the 2^+_1 state being dominated by *pf*-shell configurations [8]. Combining this observation with previous work on low-lying states in ^{32}Si from 1972 to 1998 [24–26], which suggests that the low-energy structure of ^{32}Si is described well using purely *sd*-shell orbitals, led Heery *et al.* to note that "^{32}Si therefore stands out as an outlier in B(E2; $0^+_1 \rightarrow 2^+_1$) values when compared to appropriate shell-model calculations" [27]. The B(E2; $2^+_1 \rightarrow 0^+_{\text{g.s.}}$) transition strength measured by Heery *et al.* using Coulomb excitation (Coulex) was lower than shell model calculations using the FSU [21], SDPF-MU [22], and USDB [28] calculations. This reduced transition strength is consistent with a naive picture where the 2^+_1 state has greater *pf*-shell occupancy than predicted using these shell model interactions, and thus lower overlap with the wavefunction of the $0^+_{\text{g.s.}}$ ground state. It should be noted here however that the transition strength measured by Williams *et al.* [17] using the Doppler shift attenuation method (DSAM) is consistent with the shell-model calculations, albeit at a 1.4σ tension with the Heery *et al.* Coulex measurement. *Ab initio* calculations using the VS-IMSRG many body method [29] and both $\Delta\text{NNLO}_{\text{GO}}(394)$ [30] and 1.8/2.0(EM) [31] interactions agree with the transition strength measured by Heery *et al.*

In order to resolve this discrepancy and better understand the structure of ^{32}Si and evolution of neutron orbitals approaching the $N = 20$ island of inversion, we aim to directly measure the configuration mixing of the first two 0^+ and 2^+ states using the (d, p) transfer reaction with HELIOS. Data collected in the proposed work will overlap with a similar experiment performed recently to measure states at intermediate excitation energy (see Fig. 1), but with higher statistics[1], and provide valuable information to support a future proposal aiming to constrain the $^{31}\text{Si}(n, \gamma)^{32}\text{Si}$ reaction rate. The future experiment aims to better understand the large enhancement of ^{32}S observed in pre-solar SiC grains (e.g. Ref. [32] and references therin) by measuring near-threshold states in ^{32}Si produced through the same

[1] The previous experiment ran for 3 days with a beam intensity of 2×10^4 particles per second using the same experimental setup as proposed here.

^{31}Si$(d,p)^{32}$Si reaction proposed here. The data obtained in this proposed work, along with the recent measurement at intermediate energy, will be used to better predict the near-threshold level densities in ^{32}Si and inform future experiments.

II. DESCRIPTION OF THE MEASUREMENT

In order to study the single-particle configurations in ^{32}Si, we intend to use the ^{31}Si(d,p) reaction to directly populate states in ^{32}Si. The resulting excitation energy spectrum and angular distributions will be used to extract spectroscopic factors (C^2S) for states in ^{32}Si from the ground state to $E_{\text{ex.}} \approx 6$ MeV, encompassing the first two 0^+ and 2^+ states. These data, when compared with shell model calculations, can be used to extract configurations and mixing in the low-energy structure of ^{32}Si.

The experiment will run using the ^{31}Si beam provided by ATLAS and delivered to HELIOS. According to Ref. [33], a 10 MeV/u beam of ^{31}Si, which maximizes the cross-sections at $\ell = 0$ to $\ell = 3$ momentum transfer, can be delivered at 4×10^5 pps with a purity of $\sim$5%. The beam will impinge on a standard CD_2 target of 100 μg/cm^2, giving a peak resolution of 100 keV FWHM in the excitation energy spectrum based on previous experiments [23]. The peak width is sufficient to easily distinguish between the states of interest, which are at excitation energies of $E_{\text{ex.}}$ =0, 1941.7(3), 4231.0(4), and 4984.2(11) keV for the $0^+_{\text{g.s.}}$, 2^+_1, 2^+_2, and 0^+_2 states respectively. There is no expected target degradation at beam intensities of $\sim 10^5$ pps over the length of the proposed experiment.

The recoiling ^{32}Si ions will be detected by an array of silicon detectors placed 800 cm downstream. Unscattered beam, and ^{32}Si ions with small opening angles, will be blocked by a 1.2 cm diameter beam stop placed along the beam path. A 35 cm-long, position sensitive silicon array will be placed 250 mm upstream of the target position, covering $z \in \{-600 \text{ mm}, -250 \text{ mm}\}$. Using a magnetic field of 2.85 T from HELIOS and considering excited states with $E_{\text{ex.}}$ from 0 to 6 MeV, this corresponds to centre of mass proton emission angles of 10° to 40°. Figure 2 shows the experimental setup and calculated kinematics of the reaction for various excitation energies, and Fig. 3 shows simulated spectra using this experimental setup.

III. BEAM TIME REQUIRED

For this measurement, we are requesting three days of ^{31}Si beam delivered at 4×10^5 particle per second with an energy of 10 MeV/u (310 MeV). Time allotted for beam tuning is not included in this request.

With this beamtime, we estimate that we will be able to precisely study the mixing between the two 2^+ states, as each of the corresponding peaks in the excitation spectrum should have statistical uncertainty on the order of a few percent. Rate estimates using FSU calculations [21] and cross-section estimates from previous work [23] suggest that over three days of beam time we will collect $\sim$2800 counts in each of the 2^+ peaks of the excitation spectrum, enough to extract precise configurations and differentiate between theoretical models. Additionally, $\sim$1500 and 6 counts are expected in peaks corresponding to the 0^+ states, providing an avenue to place limits on the mixing between the two 0^+ states.

IV. FIGURES

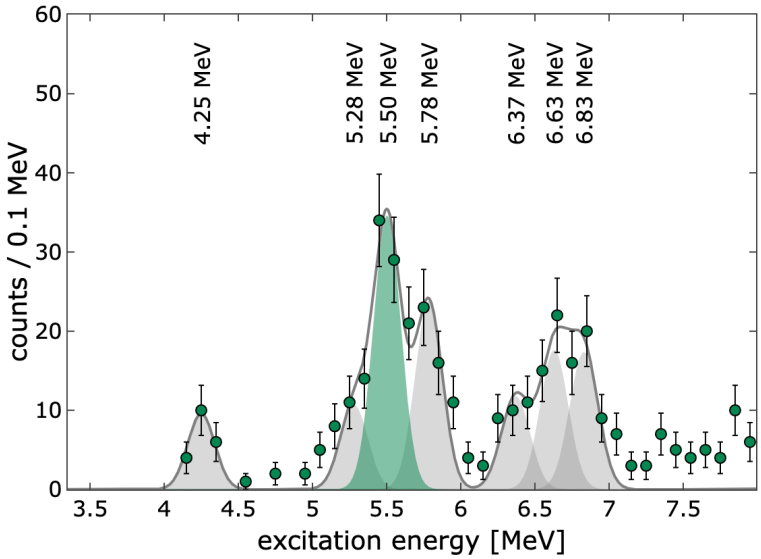

FIG. 1: Preliminary excitation energy spectrum from ^{31}Si$(d,p)^{32}$Si at intermediate excitation energy performed with HELIOS at ATLAS.

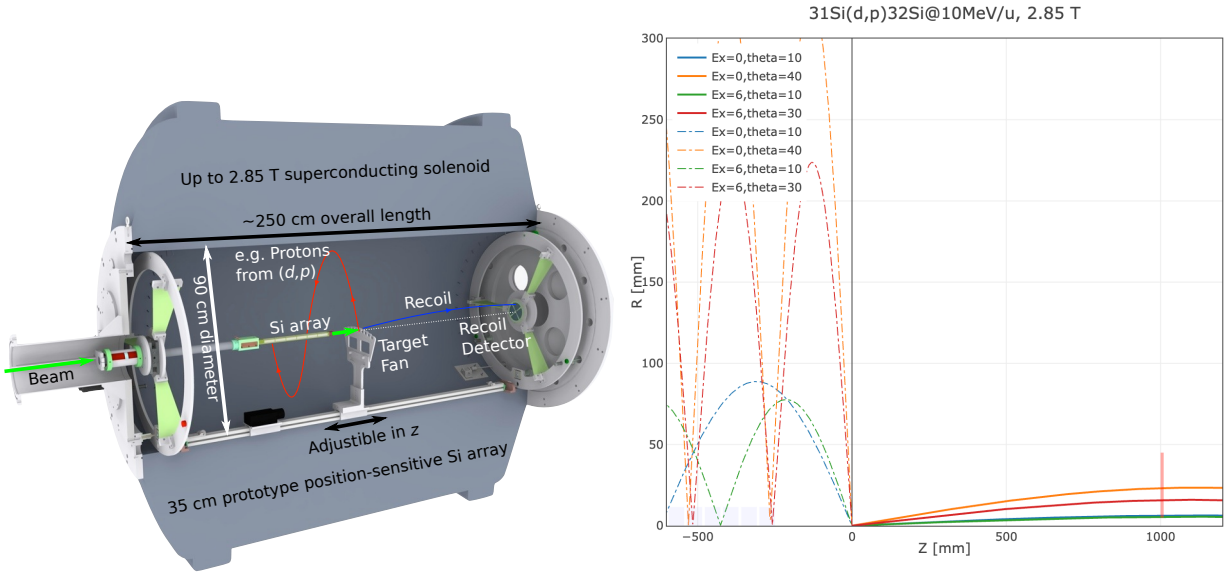

FIG. 2: (Left) Simulated image of experimental setup in HELIOS. (Right) Calculated kinematics of ^{31}Si$(d,p)^{32}$Si with the proposed experimental setup. Protons shown with dashed lines interacting with the Si array upstream of the target position, and solid lines are ^{32}Si recoils interacting with the downstream recoil detectors.

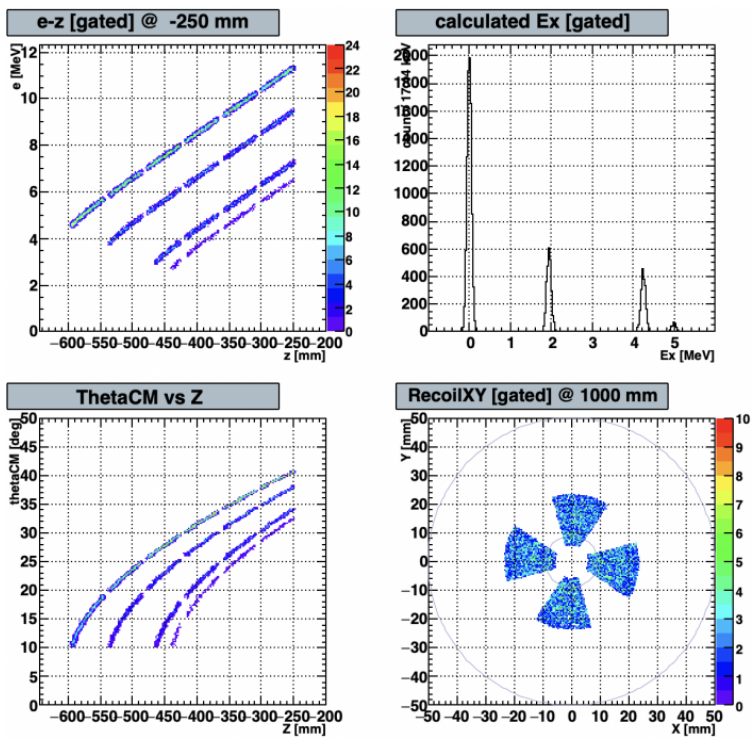

FIG. 3: Simulated spectra using the experimental setup shown in Fig. 2. All four peaks of interest are visible in the excitation energy spectrum (upper right).

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